Package mcst: Sample Scenario Illustrations

Practical Examples of Simulation, Stress Testing, and Results Extraction

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April 10, 2026

Contents

1 Installation of package **3**

2 Data Preparation **3**
2.1 Portfolio Assets . . . 3
2.2 Exogenous Instruments . . . 4
2.3 Combine and Filter . . . 5

3 Scenario 1: Base Case (Gaussian, Equal Weights) **6**
3.1 Fitted Marginal Distributions . . . 6
3.2 Simulated Portfolio Return Distribution . . . 7
3.3 Risk Metrics . . . 8
3.4 Per-Asset Empirical Summary . . . 8

4 Scenario 2: Fat Tails with t-Copula and EWMA Covariance **10**
4.1 Tail Dependence and Copula Diagnostics . . . 10
4.2 KS Uniformity Test . . . 12

5 Scenario 3: Exogenous Volatility Shock - SPY Vol x2 (GJR-GARCH) **13**
5.1 GJR-GARCH Conditional Volatility Paths . . . 13
5.2 Exogenous Propagation Channel . . . 16
5.3 Portfolio Distribution: Pre vs. Post Stress . . . 18
5.4 Stress Impact Summary . . . 19

6 Scenario 4: Oil Spike and Market Selloff (Exogenous Return and Vol Shocks) **20**
6.1 Propagation of the Compound Exogenous Shock . . . 20
6.2 Portfolio Distribution Comparison . . . 22
6.3 Stress Impact Summary . . . 23
6.4 Exogenous Historical Correlations . . . 23

7 Scenario 5: Semiconductor Selloff with SPY Offset (Mixed Endo and Exog) **24**
7.1 Mean Shift Decomposition (Waterfall) . . . 24
7.2 Portfolio Distribution Comparison . . . 26
7.3 Stress Impact Summary . . . 28

8 Portfolio Risk Decomposition **29**

9 Accessing Simulation Draws Directly **32**
9.1 NVDA: Pre vs. Post Stress Simulated Distribution 32

10 Quick Reference: Result Object Fields **33**

11 Sample Function Output: Console Printout **35**

1 Installation of package

The latest development version of the package can be installed using `remotes::install_github("calebong/mcst" dependencies = TRUE)`

```
require(remotes)
remotes::install_github("calebong/mcst", dependencies = TRUE)
```

2 Data Preparation

2.1 Portfolio Assets

For illustration we use the following packages to download 10 stocks from Yahoo finance at a daily frequency. The returns are then aggregated to weekly returns. These form the **endogenous** assets - the holdings to be simulated.

```
# simple function to convert class xts to dataframe
xts_to_dataframe <- function(xts_obj, column_names = NULL) {
  dates <- zoo::index(xts_obj)
  df    <- as.data.frame(xts_obj)
  df    <- cbind(date = as.Date(dates), df)
  if (!is.null(column_names)) names(df)[-1] <- column_names
  rownames(df) <- NULL
  df
}
```

```
# load in the libraries to get sample stocks data
library(tidyverse)
library(quantmod)
library(lubridate)

tickers <- c("NVDA","TSM","JPM","AXP","LMT","BA","UAL","XOM","NGG","PG")

out <- map(tickers, function(.x) {
  getSymbols(.x, from = "2010-01-01", to = Sys.Date(),
             src = "yahoo", auto.assign = FALSE) %>%
    xts_to_dataframe() %>%
    dplyr::select(1, 7) %>% #quickly select date and adjusted prices
    rename(Price = 2)
})
names(out) <- tickers

out_wide <- out %>%
  enframe() %>%
  unnest(cols = c(value)) %>%
  pivot_wider(names_from = "name", values_from = "Price")

weekly_returns <- out_wide %>%
  pivot_longer(cols = -date, names_to = "sector", values_to = "price") %>%
```

```r
  mutate(week = floor_date(date, unit = "week")) %>%
  group_by(sector, week) %>%
  summarise(price_week_end = last(price), .groups = "drop") %>%
  group_by(sector) %>%
  arrange(week) %>%
  mutate(weekly_return = price_week_end / lag(price_week_end) - 1) %>%
  ungroup() %>%
  filter(!is.na(weekly_return)) %>%
  dplyr::select(week, sector, weekly_return) %>%
  pivot_wider(id_cols = week, names_from = sector, values_from = weekly_return)
```

2.2 Exogenous Instruments

SPY (S&P 500 ETF - quick proxy for S&P 500 Index) and Crude Oil futures (`CL=F`) are downloaded separately as **exogenous** instruments. These carry zero portfolio weight and are used only as shock transmission channels in the stress-testing step.

```r
exo    <- c("SPY", "CL=F")
naming <- c("SPY", "Crude")

out_exo <- map(exo, function(.x) {
  getSymbols(.x, from = "2010-01-01", to = Sys.Date(),
             src = "yahoo", auto.assign = FALSE) %>%
    xts_to_dataframe() %>%
    dplyr::select(1, 7) %>% # quickly select date and adjusted prices
    rename(Price = 2)
})
names(out_exo) <- naming

out_wide_exo <- out_exo %>%
  enframe() %>%
  unnest(cols = c(value)) %>%
  pivot_wider(names_from = "name", values_from = "Price")

weekly_returns_exo <- out_wide_exo %>%
  pivot_longer(cols = -date, names_to = "sector", values_to = "price") %>%
  mutate(week = floor_date(date, unit = "week")) %>%
  group_by(sector, week) %>%
  summarise(price_week_end = last(price), .groups = "drop") %>%
  group_by(sector) %>%
  arrange(week) %>%
  mutate(weekly_return = price_week_end / lag(price_week_end) - 1) %>%
  ungroup() %>%
  filter(!is.na(weekly_return)) %>%
  dplyr::select(week, sector, weekly_return) %>%
  pivot_wider(id_cols = week, names_from = sector, values_from = weekly_return) %>%
  mutate(Crude = ifelse(Crude <= -1, -0.5, Crude))
```

2.3 Combine and Filter

To produce a data frame quickly for illustration, portfolio and exogenous returns are joined on the week key and filtered to start from 2016, giving a cleaner estimation sample.

```
comb <- weekly_returns %>%
  full_join(weekly_returns_exo, by = "week") %>%
  mutate(across(where(is.numeric), ~replace_na(., 0))) %>%
  filter(week >= "2016-01-01")

returns_data <- comb %>% dplyr::select(-Crude, -SPY)
exo_data     <- comb %>% dplyr::select(week, Crude, SPY)

cat(sprintf("Portfolio : %d assets, %d weekly obs\n",
            ncol(returns_data) - 1, nrow(returns_data)))
#> Portfolio : 10 assets, 536 weekly obs
cat(sprintf("Exogenous : %d instruments\n", ncol(exo_data) - 1))
#> Exogenous : 2 instruments

# data structure of input data frames
head(returns_data)
#> # A tibble: 6 x 11
#>   week            AXP       BA      JPM      LMT      NGG    NVDA       PG     TSM     UAL
#>   <date>        <dbl>    <dbl>    <dbl>    <dbl>    <dbl>   <dbl>    <dbl>   <dbl>   <dbl>
#> 1 2016-01-03 -0.0811  -0.101   -0.102   -0.0195  -0.0168  -0.101  -0.0433  -0.0853 -0.0944
#> 2 2016-01-10 -0.0113  -0.0335  -0.0319   0.0145  -0.0212  -0.0850 -0.0130   0.0139 -0.120
#> 3 2016-01-17 -0.125   -0.00812 -0.00158 -0.0183   0.0227   0.0494  0.0407   0.0393  0.0230
#> 4 2016-01-24 -0.0283  -0.0360   0.0448  -0.00491  0.0346   0.0295  0.0560   0.0192  0.0334
#> 5 2016-01-31  0.00897  0.0202  -0.0294   0.00446 -0.00946 -0.0976 -0.00600  0.0179 -0.0294
#> 6 2016-02-07 -0.0245  -0.105   -0.00450 -0.0276  -0.0204  -0.0265 -0.00259 -0.0224  0.0318
head(exo_data)
#> # A tibble: 6 x 3
#>   week         Crude      SPY
#>   <date>       <dbl>    <dbl>
#> 1 2016-01-03 -0.105  -0.0586
#> 2 2016-01-10 -0.113  -0.0214
#> 3 2016-01-17  0       0
#> 4 2016-01-24  0       0
#> 5 2016-01-31 -0.0812 -0.0298
#> 6 2016-02-07 -0.0469 -0.00702
```

3 Scenario 1: Base Case (Gaussian, Equal Weights)

The simplest possible call. No arguments are overridden: the function uses a multivariate Normal distribution, historical sample covariance, and equal weights across all ten assets. No stress is applied.

```
library(mcst)

result_1 <- portfolio_risk_simulation(
  historical_returns = returns_data
)
```

3.1 Fitted Marginal Distributions

Each panel shows the empirical return histogram with the fitted Normal density overlaid.

```
result_1$plots$portfolio_diagnostics$asset_distributions
```

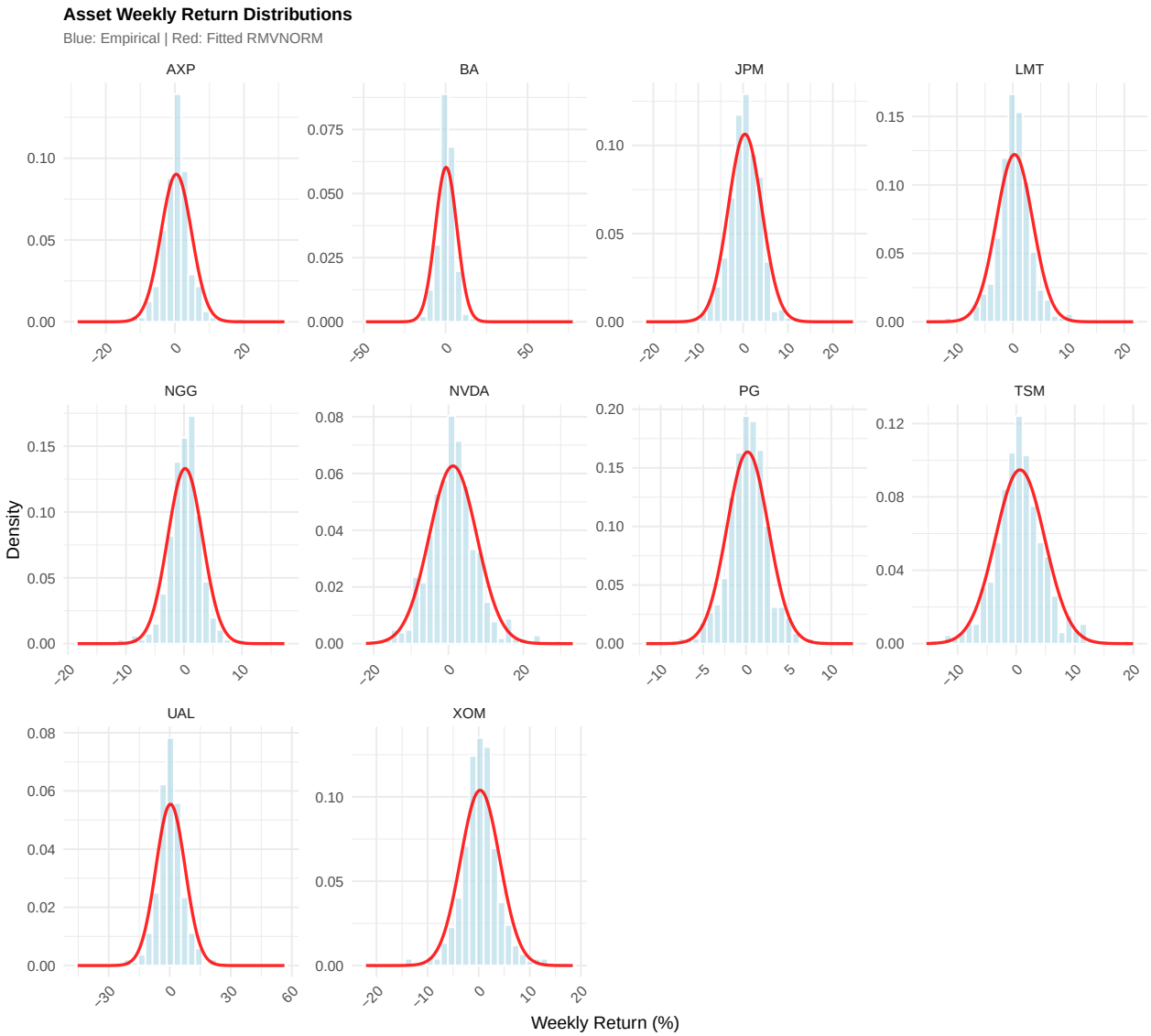

Figure 1: Scenario 1: Fitted Normal marginals vs. empirical returns per asset.

3.2 Simulated Portfolio Return Distribution

```
result_1$plots$portfolio_diagnostics$portfolio_distribution
```

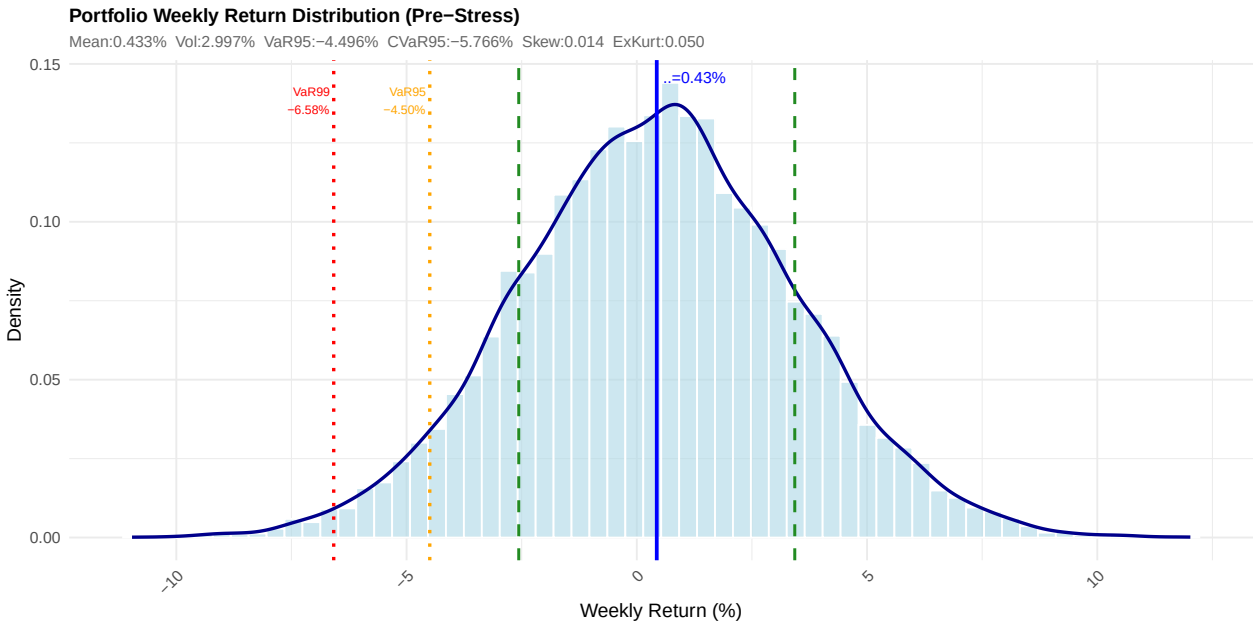

Figure 2: Scenario 1: Simulated portfolio return distribution (Gaussian, no stress).

3.3 Risk Metrics

Table 1: Scenario 1: Portfolio risk statistics (Gaussian base case).

Metric	Pre-Stress	Post-Stress
Mean (%) [primary]	0.4333	NA
Median (%)	0.4509	NA
Volatility (%)	2.9970	NA
Skewness	0.0140	NA
Excess Kurtosis	0.0498	NA
Sharpe (Annual)	1.0425	NA
Annual Return (%)	25.2082	NA
Annual Vol (%)	21.6119	NA
VaR 90% (%)	-3.3602	NA
VaR 95% (%)	-4.4964	NA
VaR 99% (%)	-6.5820	NA
CVaR 90% (%)	-4.8202	NA
CVaR 95% (%)	-5.7662	NA
CVaR 99% (%)	-7.5908	NA

3.4 Per-Asset Empirical Summary

Table 2: Scenario 1: Per-asset empirical return statistics.

Asset	Mean	SD	Skewness	Kurtosis	Min	Max
AXP	0.0041	0.0442	0.1526	10.1695	-0.2558	0.2883
BA	0.0031	0.0663	1.7006	33.5329	-0.4418	0.7051
JPM	0.0041	0.0375	-0.0044	7.3702	-0.1964	0.2226
LMT	0.0030	0.0327	0.2890	8.2489	-0.1409	0.1963
NGG	0.0020	0.0300	-0.5157	8.6456	-0.1667	0.1575

NVDA	0.0122	0.0637	0.3594	4.4950	-0.2005	0.3019
PG	0.0020	0.0244	-0.1844	5.7266	-0.1063	0.1140
TSM	0.0066	0.0421	0.2182	4.1879	-0.1393	0.1817
UAL	0.0036	0.0720	0.6811	12.2323	-0.4116	0.5125
XOM	0.0029	0.0384	-0.1046	6.6570	-0.2007	0.1674

4 Scenario 2: Fat Tails with t-Copula and EWMA Covariance

Changes from base case: Multivariate t returns (`rmvt`), t-copula to capture tail co-dependence, EWMA covariance ($\lambda = 0.97$) so recent observations receive higher weight, and per-asset degrees-of-freedom calibrated from the data (`calibration_method = "full"`).

```
result_2 <- portfolio_risk_simulation(
  historical_returns  = returns_data,
  sim_returns_dist    = "rmvt",
  cov_estimation      = "ewma",
  cov_lambda          = 0.97,
  calibrate_params    = TRUE,
  calibration_method  = "full",
  copula              = TRUE,
  copula_type         = "t",
  copula_df           = 4,
  n_sim_returns       = NULL,
  propagation         = TRUE,
  exogenous_returns   = NULL,
  exogenous_shock     = NULL,
  exogenous_volatility_shock = NULL,
  df                  = 4,
  seed                = 123
)
```

4.1 Tail Dependence and Copula Diagnostics

The left panel shows the pairwise lower tail dependence coefficient λ - the probability that two assets move to extreme values simultaneously. The right panel compares fitted copula correlations to the Kendall τ-implied starting values: points far from the diagonal indicate that MLE optimisation shifted that pair's correlation materially from the method-of-moments estimate.

```
result_2$plots$portfolio_diagnostics$tail_dependency
```

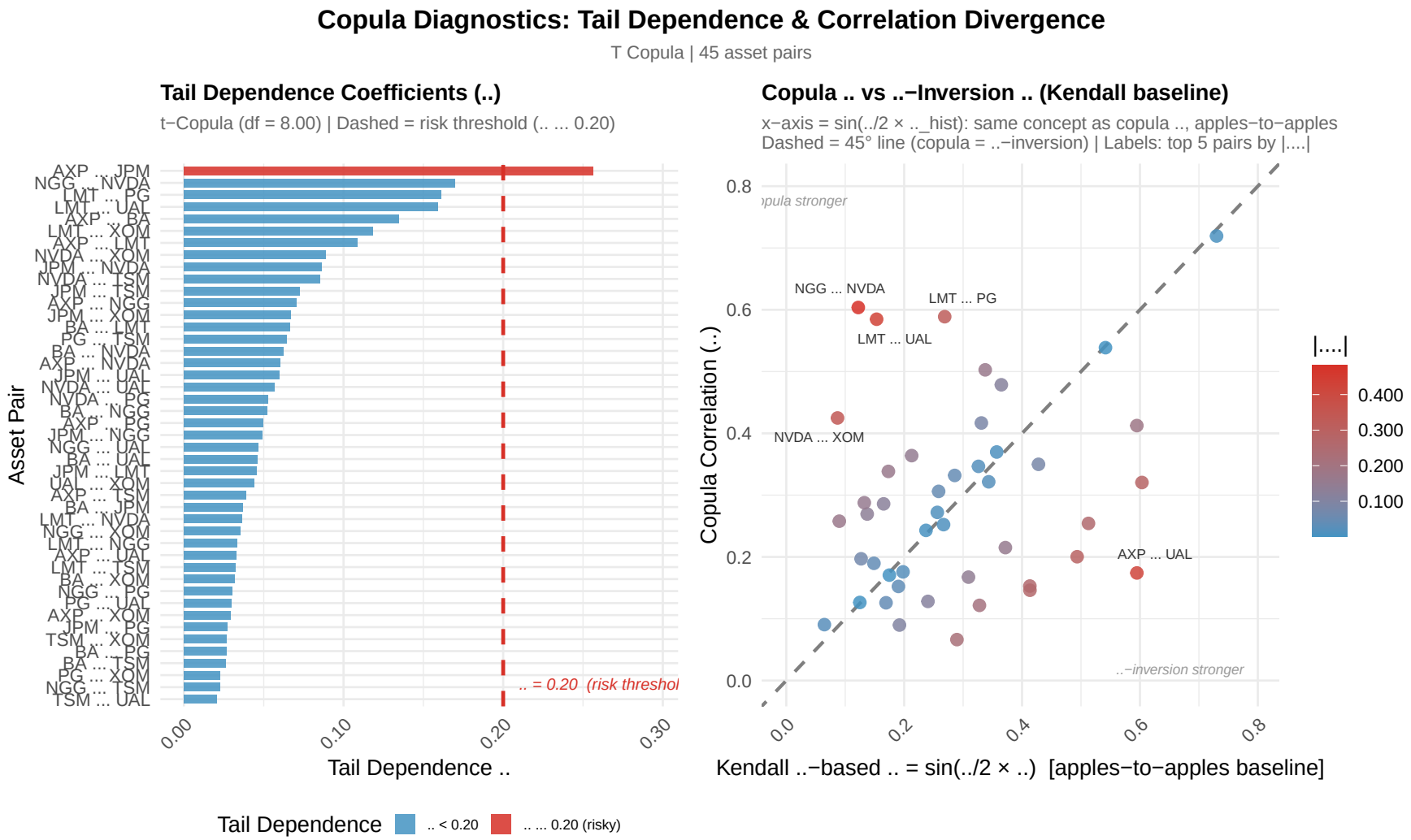

Figure 3: Scenario 2: Pairwise tail dependence and copula vs. Kendall baseline.

4.2 KS Uniformity Test

The KS test checks whether each asset's PIT residuals are Uniform[0,1] after the t-marginal is fitted. A rejection ($p < 0.05$) flags a misspecified marginal for that asset.

Table 3: Scenario 2: KS uniformity test on PIT residuals.

Asset	KS_stat	p_value	Flag
AXP	0.0660	0.0188	**REJECT**
BA	0.0976	0.0001	**REJECT**
JPM	0.0413	0.3187	**OK** |
LMT	0.0590	0.0479	**REJECT**
NGG	0.0516	0.1156	**OK** |
NVDA	0.0343	0.5553	**OK** |
PG	0.0496	0.1436	**OK** |
TSM	0.0359	0.4930	**OK** |
UAL	0.0698	0.0108	**REJECT**
XOM	0.0626	0.0300	**REJECT**

5 Scenario 3: Exogenous Volatility Shock - SPY Vol x2 (GJR-GARCH)

Shock: SPY volatility doubles ($k_{SPY} = 2$); Crude is unchanged ($k_{Crude} = 1$). No return shock is applied, so **portfolio means are unaffected** - only the covariance widens via the Law of Total Variance addon. GJR-GARCH estimates per-asset forward volatility, capturing volatility clustering and the leverage effect.

```
exogenous_volatility_shock <- list(SPY = 2, Crude = 1)
```

```
result_3 <- portfolio_risk_simulation(
  historical_returns          = returns_data,
  sim_returns_dist            = "rmvt",
  cov_estimation              = "garch",
  calibrate_params            = TRUE,
  calibration_method          = "full",
  copula                      = TRUE,
  copula_type                 = "t",
  copula_df                   = 4,
  n_sim_returns               = 10000,
  propagation                 = TRUE,
  exogenous_returns           = exo_data,
  exogenous_shock             = NULL,
  exogenous_volatility_shock  = exogenous_volatility_shock,
  df                          = 4,
  seed                        = 123
)
```

5.1 GJR-GARCH Conditional Volatility Paths

Each panel shows: salmon bars = raw returns; blue line = in-sample conditional $\hat{\sigma}_t$; red dashed = long-run unconditional $\bar{\sigma}$; green diamond = 1-step-ahead forecast. Assets where the forecast sits well above $\bar{\sigma}$ are in an elevated-volatility regime.

```r
result_3$plots$portfolio_diagnostics$time_series
```

Figure 4: Scenario 3: GJR-GARCH conditional volatility paths per asset.

5.2 Exogenous Propagation Channel

The beta $\beta_{pe} = \mathrm{Cov}(X_p, X_e)/\mathrm{Var}(X_e)$ quantifies the transmission of SPY volatility to each portfolio asset. With a pure vol shock (no return shock), these betas determine the size of the LTV covariance addon - not any mean shift.

```
result_3$stressed_plots$exogenous_propagation
```

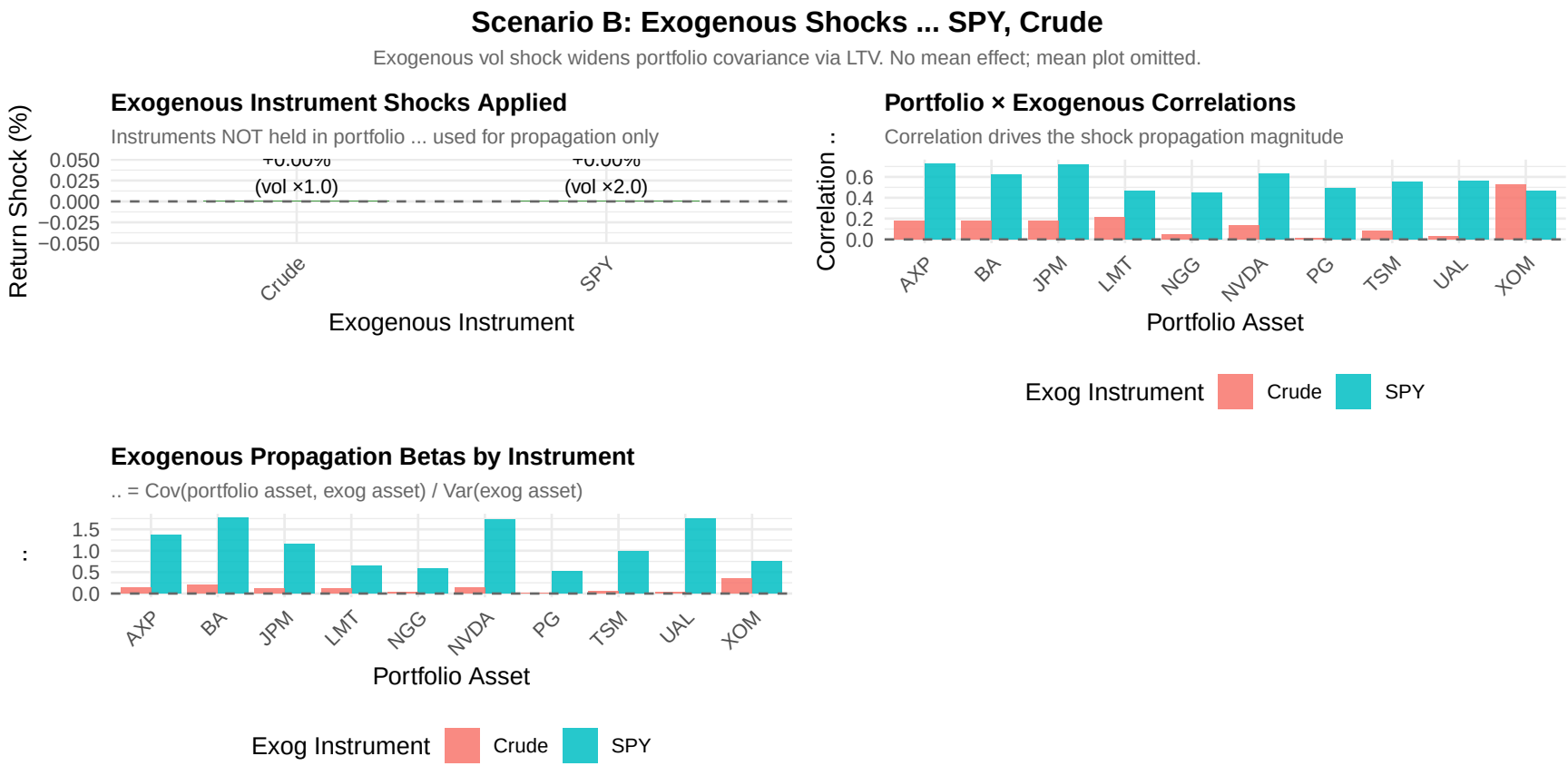

Figure 5: Scenario 3: SPY propagation betas per portfolio asset.

5.3 Portfolio Distribution: Pre vs. Post Stress

```
result_3$stressed_plots$portfolio_distribution_comparison
```

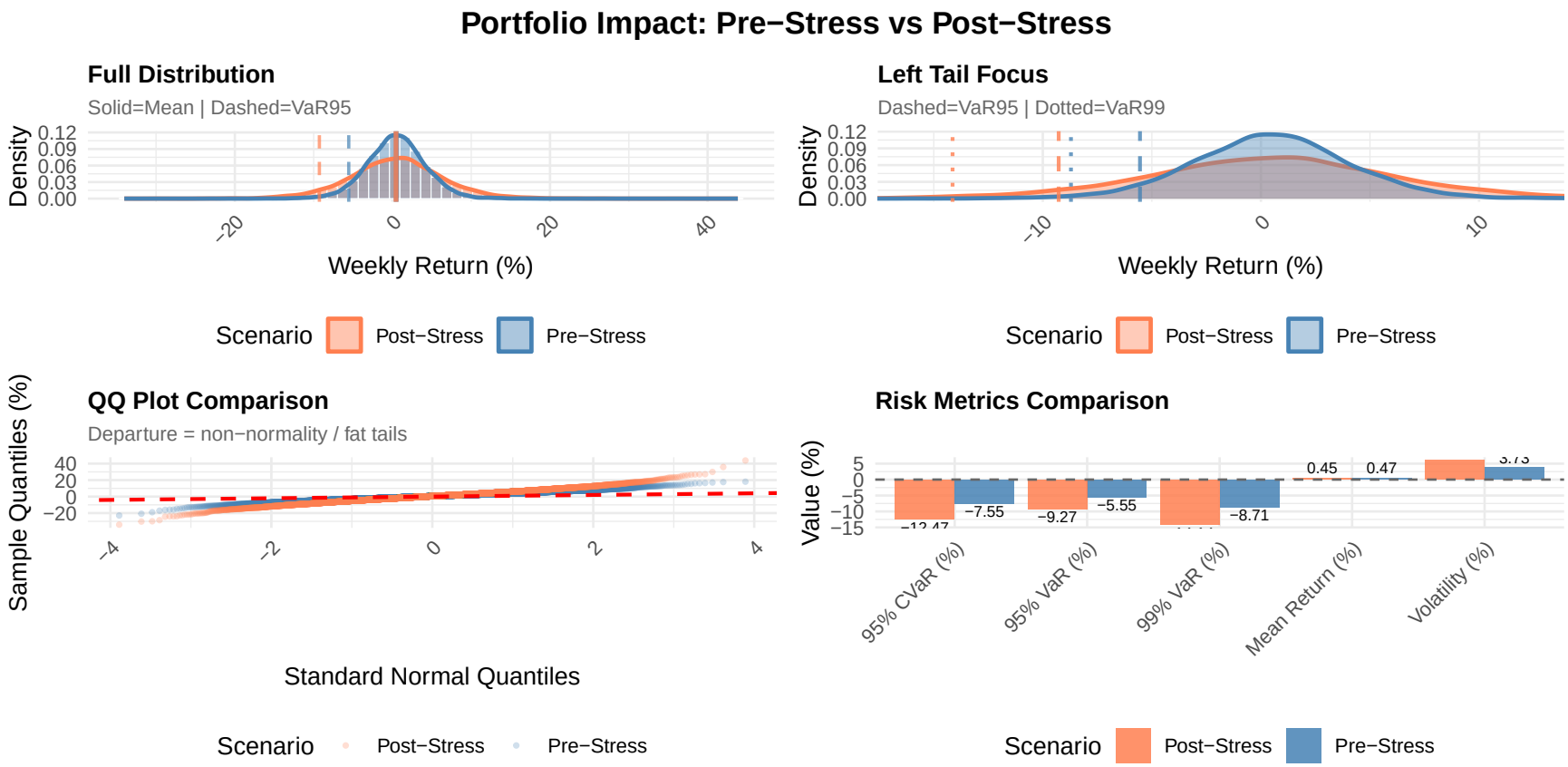

Figure 6: Scenario 3: Portfolio return distribution, base vs. SPY vol x2.

5.4 Stress Impact Summary

Table 4: Scenario 3: Risk metric changes from SPY volatility doubling.

Metric	Pre_Stress	Post_Stress	Change
Expected Return (%)	0.4750	0.4513	-0.0237
Volatility (%)	3.7340	5.9961	2.2621
Skewness	-0.0314	0.0516	0.0831
Excess Kurtosis	1.0978	1.4231	0.3252
VaR 95% (%)	-5.5452	-9.2729	-3.7276
VaR 99% (%)	-8.7100	-14.1382	-5.4282
CVaR 95% (%)	-7.5474	-12.4717	-4.9243

6 Scenario 4: Oil Spike and Market Selloff (Exogenous Return and Vol Shocks)

Shock: SPY falls 5% ($\delta_{SPY} = -0.05$) with vol doubled; Crude surges 20% ($\delta_{Crude} = +0.20$) with vol doubled. This may simulate a stagflationary stress scenario: energy costs surge while equity markets sell off simultaneously.

Both return shocks enter the Schur complement together. The shocked block is $s = \{SPY, Crude\}$ and all portfolio asset means are updated by $\mu_{p|s} = \mu_p + \Sigma_{pe}\Sigma_{ee}^{-1}\delta_e$. Energy-sensitive holdings (XOM) received a net positive effect, whereby a positive effect from a high beta to Crude Oil is partially offset by a negative effect from SPY declines. Broad-market holdings (JPM, AXP, NVDA) are dominated by the SPY decline.

```
exogenous_shock            <- list(SPY = -0.05, Crude = 0.20)
exogenous_volatility_shock <- list(SPY = 2,     Crude = 2)
```

```
result_4 <- portfolio_risk_simulation(
  historical_returns          = returns_data,
  sim_returns_dist            = "rmvt",
  cov_estimation              = "garch",
  calibrate_params            = TRUE,
  calibration_method          = "full",
  copula                      = TRUE,
  copula_type                 = "t",
  copula_df                   = 4,
  n_sim_returns               = NULL,
  propagation                 = TRUE,
  exogenous_returns           = exo_data,
  exogenous_shock             = exogenous_shock,
  exogenous_volatility_shock  = exogenous_volatility_shock,
  df                          = 4,
  seed                        = 123
)
```

6.1 Propagation of the Compound Exogenous Shock

```
result_4$stressed_plots$exogenous_propagation
```

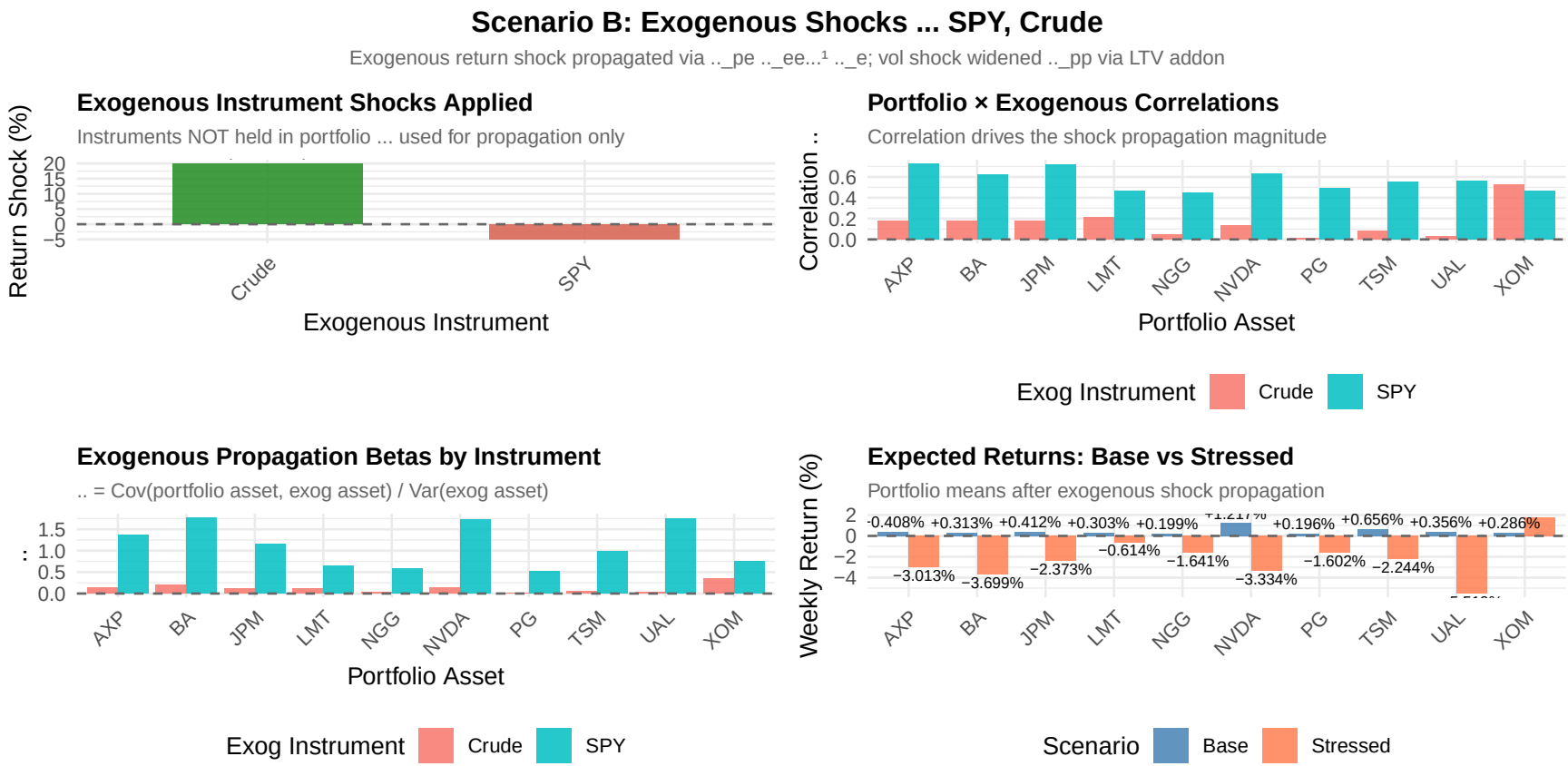

Figure 7: Scenario 4: Conditional mean shifts from SPY -5pct and Crude +20pct.

6.2 Portfolio Distribution Comparison

```
result_4$stressed_plots$portfolio_distribution_comparison
```

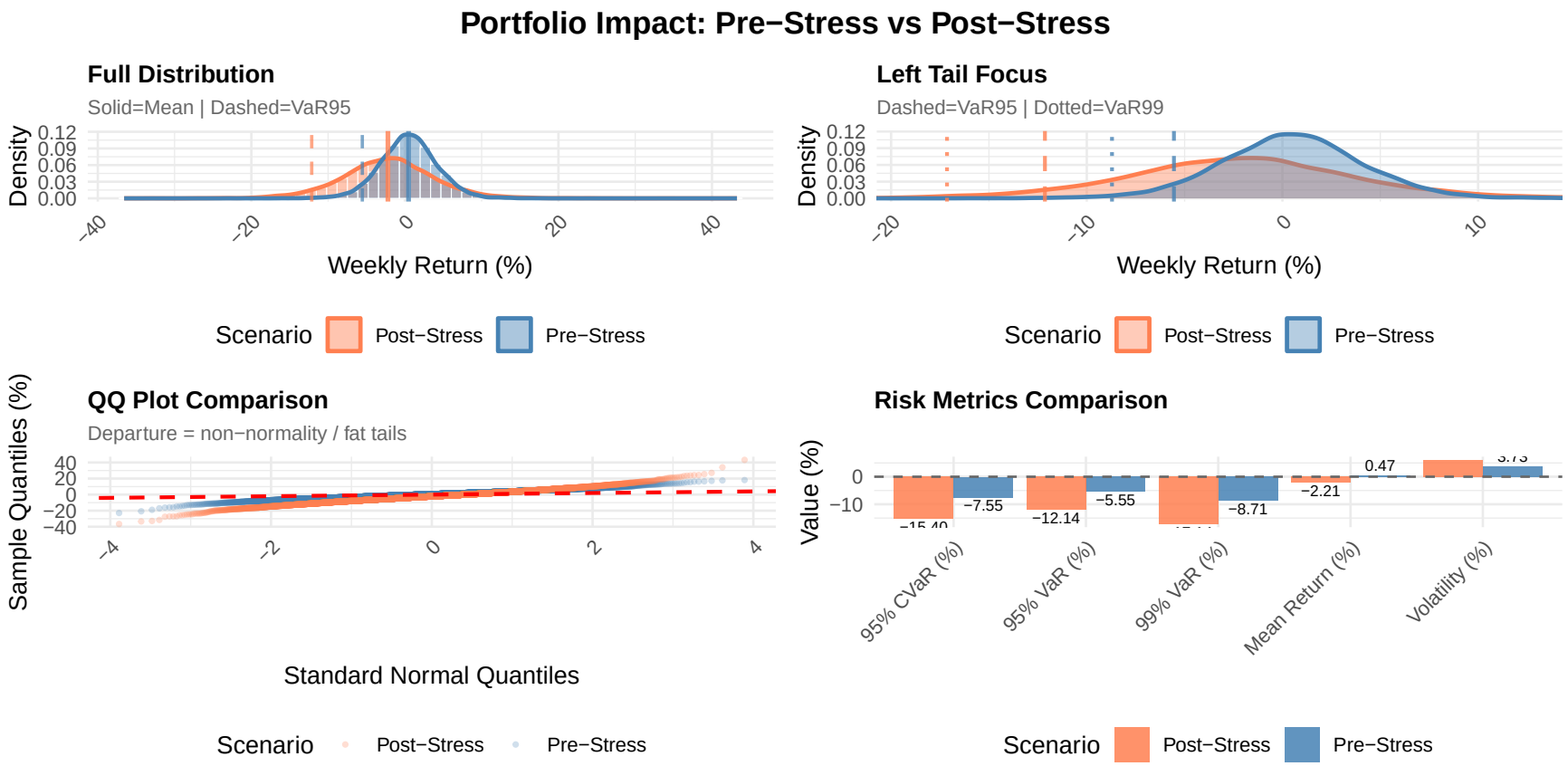

Figure 8: Scenario 4: Portfolio distribution under oil spike and market selloff.

6.3 Stress Impact Summary

Table 5: Scenario 4: Risk metrics under stagflationary shock.

Metric	Pre_Stress	Post_Stress	Change
Expected Return (%)	0.4750	-2.2075	-2.6824
Volatility (%)	3.7340	6.1161	2.3821
Skewness	-0.0314	0.0529	0.0843
Excess Kurtosis	1.0978	1.4258	0.3280
VaR 95% (%)	-5.5452	-12.1374	-6.5922
VaR 99% (%)	-8.7100	-17.1416	-8.4316
CVaR 95% (%)	-7.5474	-15.3970	-7.8496

6.4 Exogenous Historical Correlations

This table explains the propagation magnitudes above. Assets with high SPY correlation (JPM, AXP, NVDA) absorb the largest mean shift from the SPY leg; XOM's positive Crude correlation provides a partial offset.

Table 6: Scenario 4: Historical correlations, portfolio vs. SPY and Crude.

Asset	Crude	SPY
AXP	0.175	0.729
BA	0.179	0.627
JPM	0.182	0.718
LMT	0.215	0.469
NGG	0.045	0.452
NVDA	0.131	0.633
PG	0.014	0.490
TSM	0.085	0.554
UAL	0.033	0.566
XOM	0.527	0.464

7 Scenario 5: Semiconductor Selloff with SPY Offset (Mixed Endo and Exog)

Shock: NVDA and TSM each fall 5% with vol multiplied 1.5x (endogenous return and vol shocks). Simultaneously, SPY rises 2.5% with vol 1.25x (exogenous return and vol shock). This captures a sector-specific stress ie regulatory action or export controls on semiconductors that occurs against a mild broader market rally.

The Schur complement operates on a combined shocked block $s = \{NVDA, TSM, SPY\}$. Assets correlated with the semiconductor pair receive a negative propagated mean shift; the SPY tailwind partially offsets this for holdings with high market beta.

```
asset_shock                <- list(NVDA = -0.05, TSM = -0.05)
volatility_shock           <- list(NVDA = 1.5,   TSM = 1.5)
exogenous_shock            <- list(SPY = 0.025)
exogenous_volatility_shock <- list(SPY = 1.25)
```

```
result_5 <- portfolio_risk_simulation(
  historical_returns          = returns_data,
  sim_returns_dist            = "rmvt",
  cov_estimation              = "garch",
  calibrate_params            = TRUE,
  calibration_method          = "full",
  copula                      = TRUE,
  copula_type                 = "t",
  copula_df                   = 4,
  n_sim_returns               = NULL,
  propagation                 = TRUE,
  asset_shock                 = asset_shock,
  volatility_shock            = volatility_shock,
  exogenous_returns           = exo_data,
  exogenous_shock             = exogenous_shock,
  exogenous_volatility_shock  = exogenous_volatility_shock,
  df                          = 4,
  seed                        = 123
)
```

7.1 Mean Shift Decomposition (Waterfall)

The stacked bar chart decomposes each asset's total conditional mean shift into three components: (1) **direct endogenous shock** (non-zero only for NVDA and TSM), (2) **propagated from endo** (Schur from the NVDA/TSM block), and (3) **propagated from exog** (Schur from the SPY return shock). The diamond marker shows the net total.

```
result_5$stressed_plots$mixed_shock_effects
```

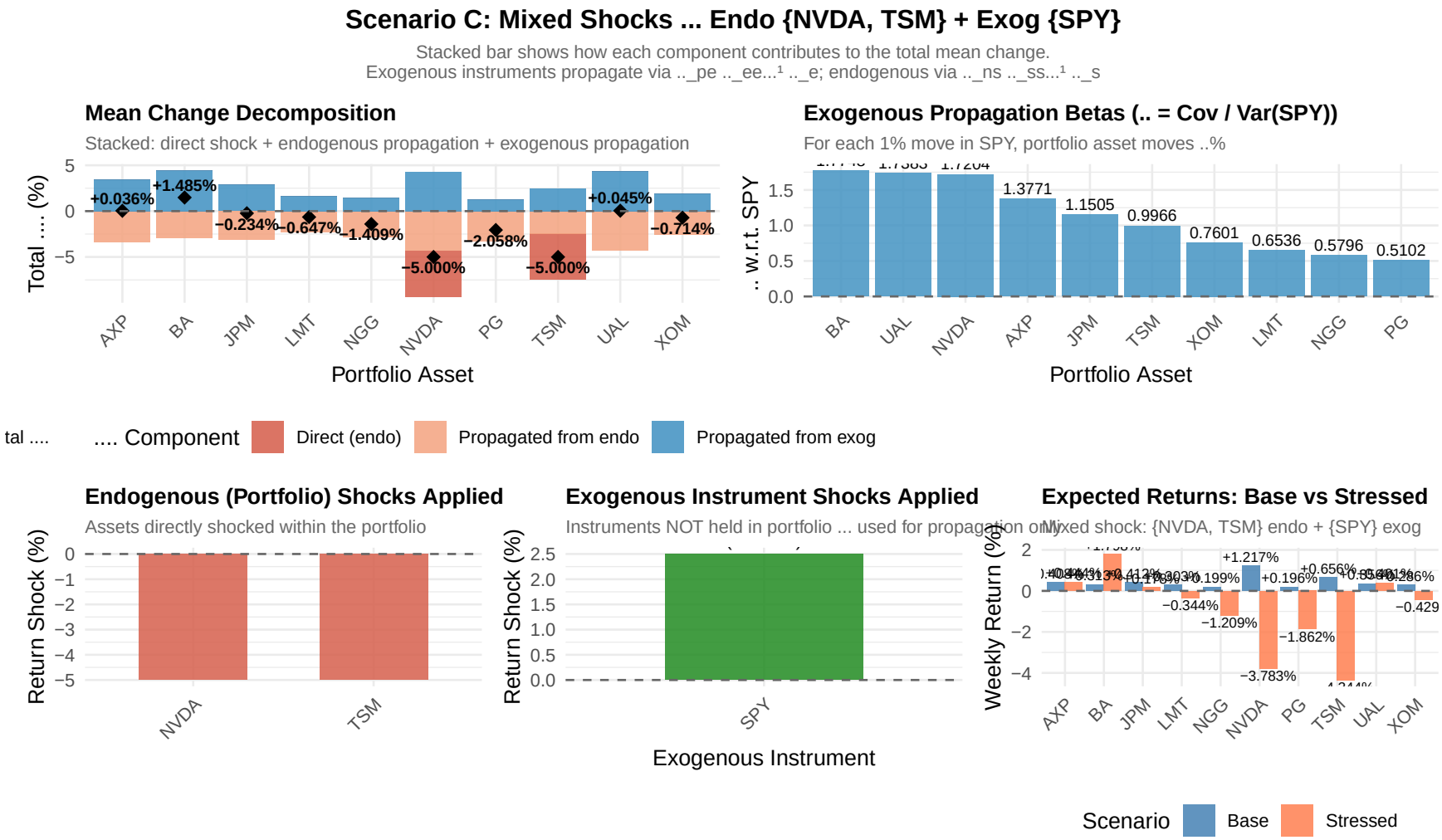

Figure 9: Scenario 5: Mean shift decomposition - direct endo, propagated endo, propagated exog.

7.2 Portfolio Distribution Comparison

Despite the SPY tailwind, the portfolio distribution shifts left because NVDA and TSM are large equal-weight components with high intra-sector correlations that propagate their decline across correlated holdings.

```
result_5$stressed_plots$portfolio_distribution_comparison
```

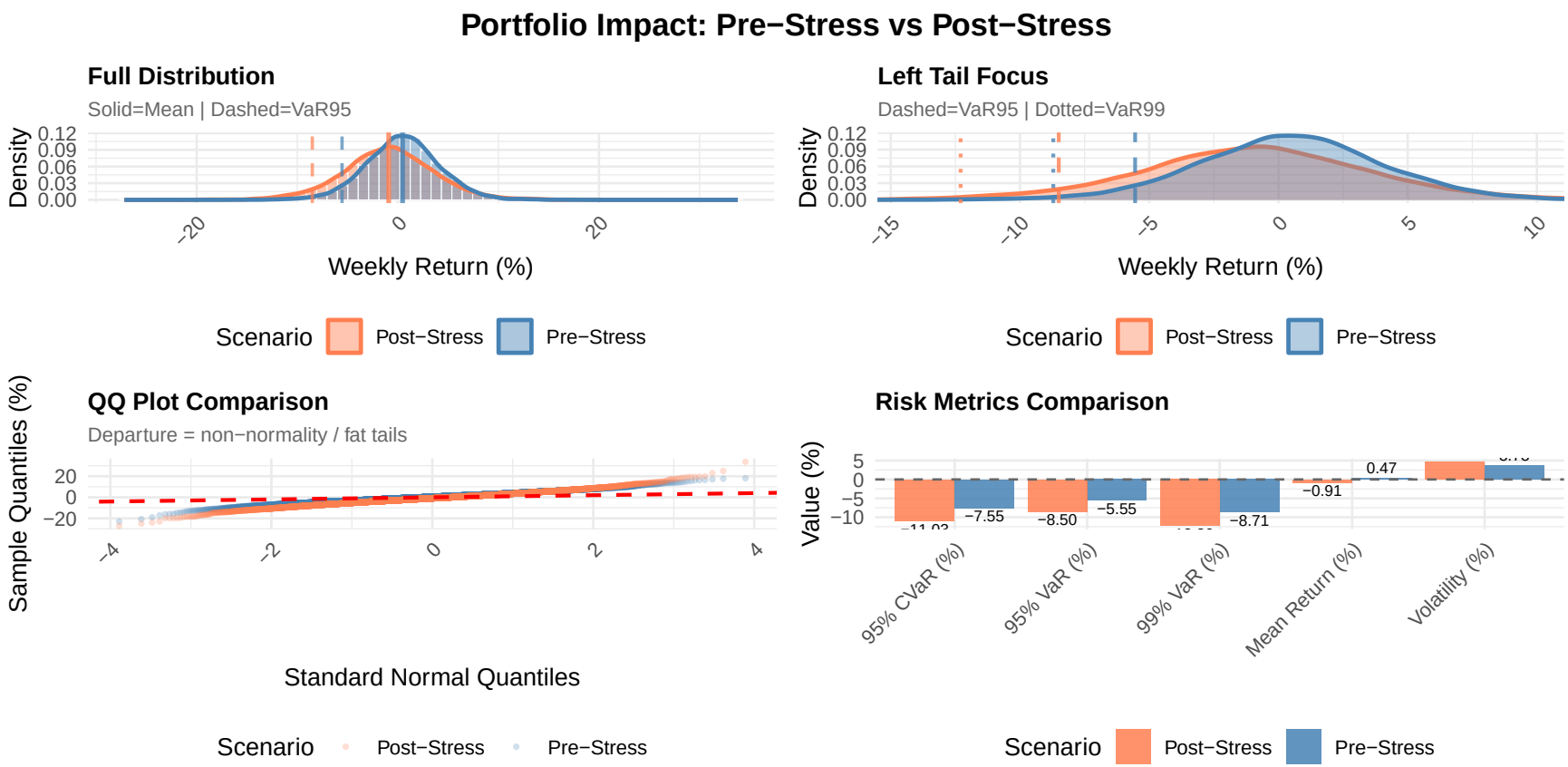

Figure 10: Scenario 5: Portfolio distribution - semiconductor selloff with SPY rally.

7.3 Stress Impact Summary

Table 7: Scenario 5: Risk metrics under semiconductor-specific stress.

Metric	Pre_Stress	Post_Stress	Change
Expected Return (%)	0.4750	-0.9053	-1.3803
Volatility (%)	3.7340	4.6844	0.9504
Skewness	-0.0314	0.0514	0.0828
Excess Kurtosis	1.0978	1.4363	0.3385
VaR 95% (%)	-5.5452	-8.4992	-2.9540
VaR 99% (%)	-8.7100	-12.2959	-3.5859
CVaR 95% (%)	-7.5474	-11.0326	-3.4852

8 Portfolio Risk Decomposition

We can also extract the contribution to total portfolio risk by calling `decompose_portfolio_risk()` on the result object:

```
decomp_result_5 <- decompose_portfolio_risk(result_5,
                                            var_method = "tail")
```

```
decomp_result_5$plots$risk_decomposition
```

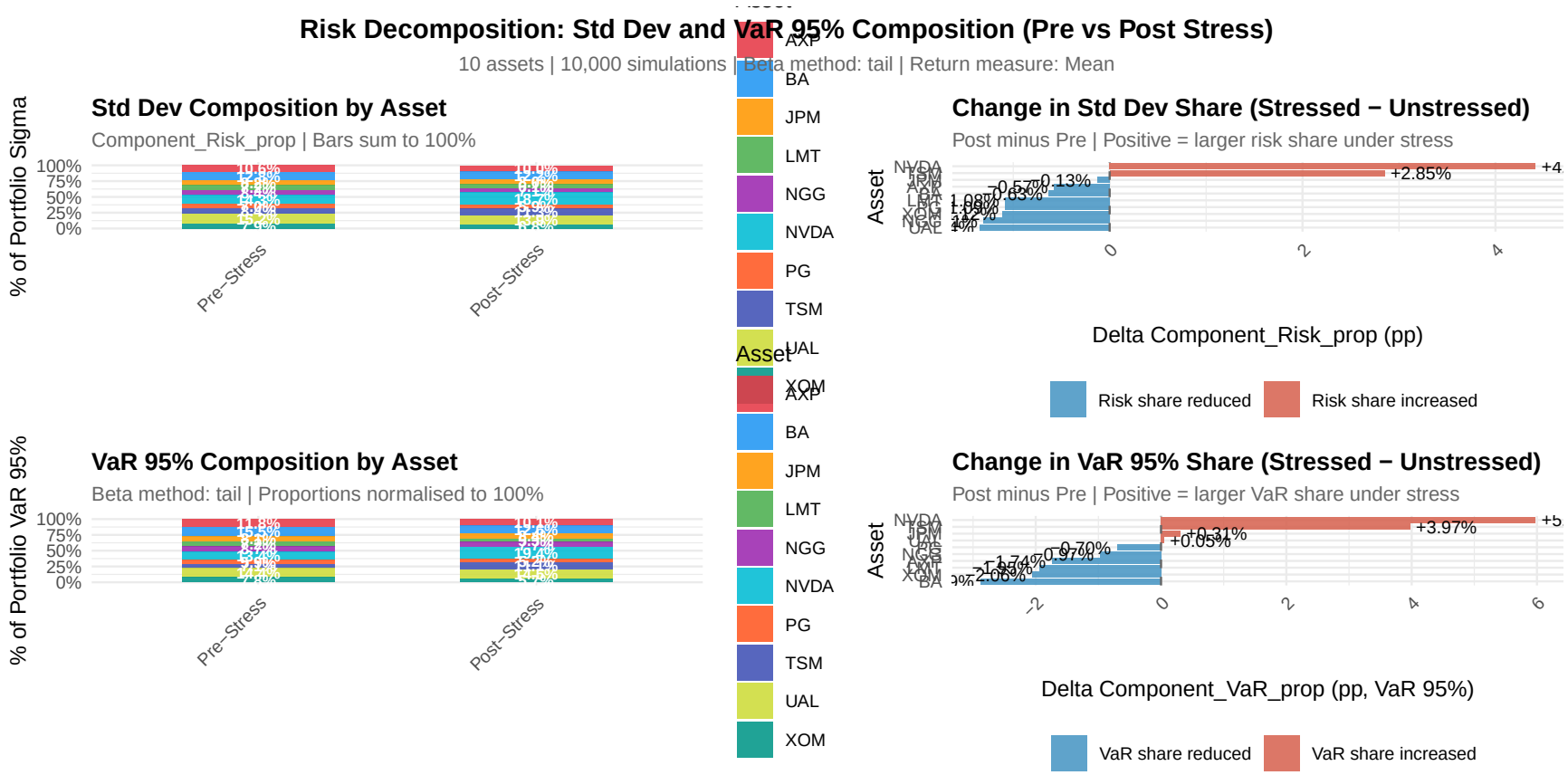

Figure 11: Scenario 5: Risk Decomposition (proportion) of non-stressed vs stressed scenarios.

```
decomp_result_5$plots$summary_canvas
```

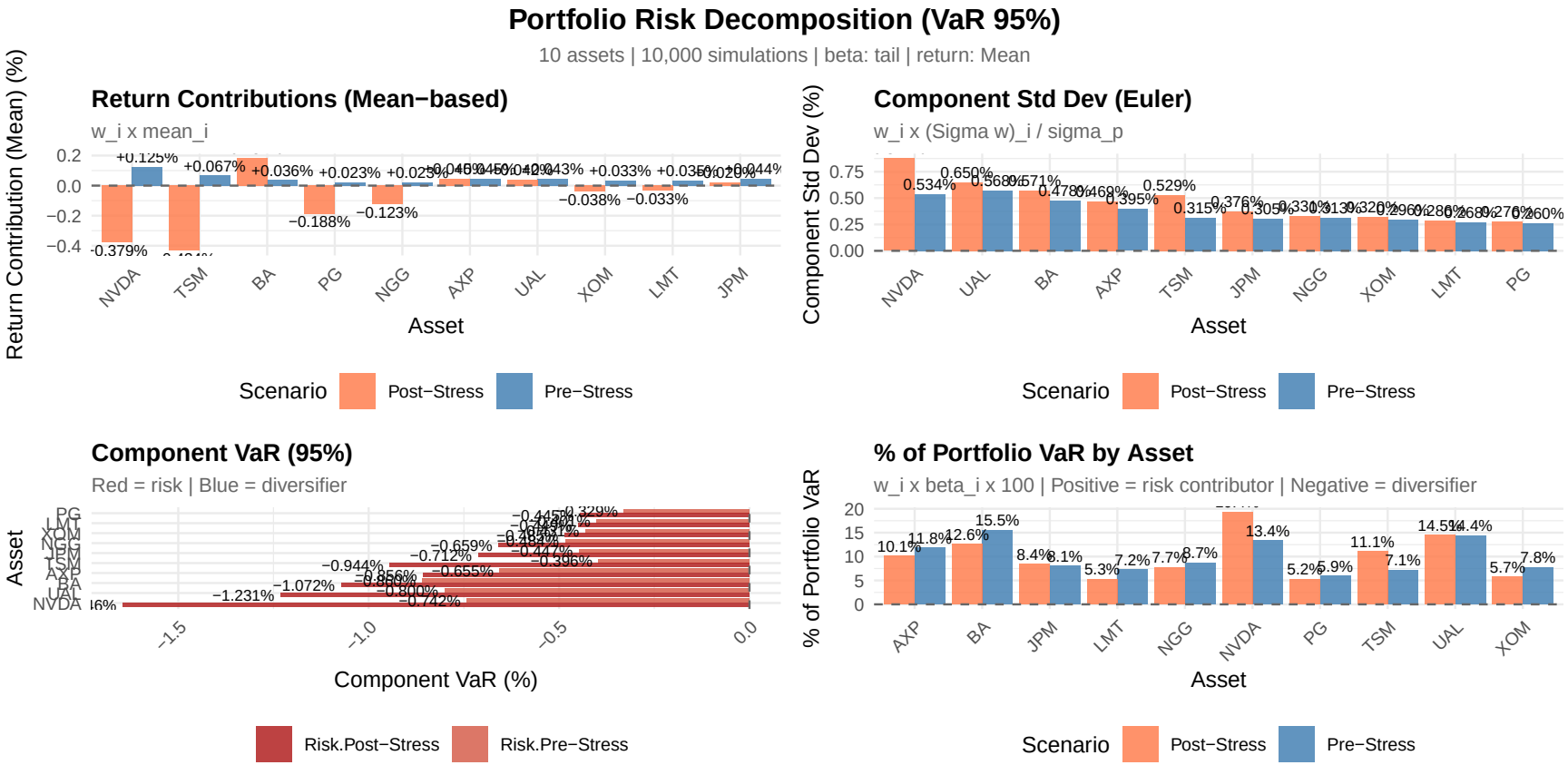

Figure 12: Scenario 5: Risk Decomposition of non-stressed vs stressed scenarios.

9 Accessing Simulation Draws Directly

The raw per-asset simulated draws are stored as $n_{\text{sim}} \times n_{\text{assets}}$ data frames - accessible without any aggregation to portfolio level. This enables custom downstream analysis such as per-asset VaR.

```
draws_pre  <- result_5$asset_draws
draws_post <- result_5$asset_draws_stress

cat("Pre-stress  draws:", nrow(draws_pre),  "rows x", ncol(draws_pre),  "cols\n")
#> Pre-stress  draws: 10000 rows x 10 cols
cat("Post-stress draws:", nrow(draws_post), "rows x", ncol(draws_post), "cols\n")
#> Post-stress draws: 10000 rows x 10 cols
cat("Columns:", paste(names(draws_pre), collapse = ", "), "\n")
#> Columns: AXP, BA, JPM, LMT, NGG, NVDA, PG, TSM, UAL, XOM
```

9.1 NVDA: Pre vs. Post Stress Simulated Distribution

```
nvda_comp <- data.frame(
  Return   = c(draws_pre$NVDA, draws_post$NVDA),
  Scenario = rep(c("Pre-Stress","Post-Stress"), each = nrow(draws_pre))
)

means_df <- nvda_comp %>%
  group_by(Scenario) %>%
  summarise(m = mean(Return * 100), .groups = "drop")

ggplot(nvda_comp, aes(x = Return * 100, fill = Scenario)) +
  geom_density(alpha = 0.45, colour = NA) +
  geom_vline(data = means_df, aes(xintercept = m, colour = Scenario),
             linetype = "dashed", linewidth = 0.8) +
  scale_fill_manual(values   = c("Pre-Stress" = "steelblue","Post-Stress" = "coral")) +
  scale_colour_manual(values = c("Pre-Stress" = "steelblue","Post-Stress" = "coral")) +
  labs(title    = "NVDA: Simulated Return Distribution, Pre vs. Post Stress",
       subtitle = "Scenario 5: NVDA direct shock -5%, vol x1.5 | SPY +2.5%, vol x1.25",
       x = "Weekly Return (%)", y = "Density", fill = NULL, colour = NULL) +
  theme_minimal(base_size = 11) +
  theme(legend.position = "bottom")
```

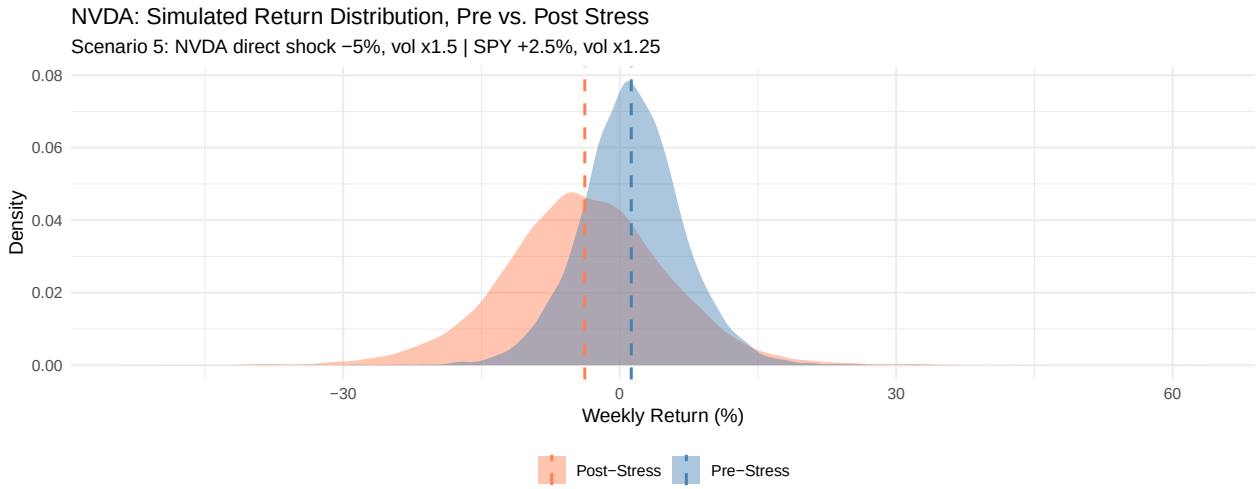

Figure 13: NVDA simulated return distribution, pre and post Scenario 5 stress.

10 Quick Reference: Result Object Fields

Table 8: Quick reference: result object fields and plot accessors.

Field	Description
result$portfolio_stats	14-row data frame: risk metrics pre and post stress
result$asset_draws	n_sim x n_assets data frame of pre-stress per-asset draws
result$asset_draws_stress	n_sim x n_assets data frame of post-stress per-asset draws
result$asset_empirical_summary	Per-asset empirical mean, SD, skewness, kurtosis, min, max
result$exogenous_historical_correlation	Portfolio x exogenous Pearson correlations (Asset + one col per instrument)
result$exogenous_asset_empirical_summary	Exogenous instrument empirical summary: mean, SD, skewness, kurtosis
result$ks_test	KS test results: Asset, KS_stat, p_value, Flag
result$stress_impact_summary	7-row data frame: Metric, Pre_Stress, Post_Stress, Change
result$stressed_covariance	n x n stressed covariance matrix after shock application
result$stressed_means	Named vector of stressed conditional means per asset
result$final_covariance	n x n final forward-looking covariance matrix
result$copula_correlation	n x n fitted copula correlation matrix
result$tail_dependence	n x n tail dependence matrix (t-copula only)
result$calibrated_params	Calibrated shape parameters per asset
result$plots$...$asset_distributions	Histogram + fitted density overlay, one panel per asset
result$plots$...$portfolio_distribution	Simulated portfolio return distribution (histogram + KDE)
result$plots$...$tail_dependency	Tail dependence bar chart + copula vs. Kendall tau scatter
result$plots$...$time_series	GJR-GARCH conditional vol time series (GARCH cov_estimation only)
result$stressed_plots$portfolio_distribution_comparison	Overlaid pre vs. post stress portfolio return distributions

Table 8: Quick reference: result object fields and plot accessors. *(continued)*

Field	Description
result$stressed_plots$exogenous_propagation	Exogenous beta and propagation channel plots (Scenario B)
result$stressed_plots$mixed_shock_effects	Waterfall decomposition: direct + propagated endo + exog (Scenario C)

11 Sample Function Output: Console Printout

The `portfolio_risk_simulation` function produces a detailed diagnostic console printout during execution. This section captures a representative example from Scenario 4 (oil spike and market selloff) to illustrate what the output looks like, covering data preprocessing, copula fitting, stress-testing steps, and the final stress impact summary.

historical_returns converted from tibble to data frame Date column detected: 'week' (536 dates, 2016-01-03 to 2026-04-05) Data frequency inferred: Weekly (52 periods/year)

==========
DATA PREPROCESSING SUMMARY ==========
Assets : 10 Asset names : AXP, BA, JPM, LMT, NGG, NVDA, PG, TSM, UAL, XOM Observations : 536 (from 536 raw rows) Distribution : rmvt Copula : YES (T) Cov estimation : GARCH

Per-asset empirical summary: AXP: mean=0.0041 sd=0.0442 skew=0.153 kurt=10.169 min=-0.2558 max=0.2883 BA: mean=0.0031 sd=0.0663 skew=1.701 kurt=33.533 min=-0.4418 max=0.7051 JPM: mean=0.0041 sd=0.0375 skew=-0.004 kurt=7.370 min=-0.1964 max=0.2226 LMT: mean=0.0030 sd=0.0327 skew=0.289 kurt=8.249 min=-0.1409 max=0.1963 NGG: mean=0.0020 sd=0.0300 skew=-0.516 kurt=8.646 min=-0.1667 max=0.1575 NVDA: mean=0.0122 sd=0.0637 skew=0.359 kurt=4.495 min=-0.2005 max=0.3019 PG: mean=0.0020 sd=0.0244 skew=-0.184 kurt=5.727 min=-0.1063 max=0.1140 TSM: mean=0.0066 sd=0.0421 skew=0.218 kurt=4.188 min=-0.1393 max=0.1817 UAL: mean=0.0036 sd=0.0720 skew=0.681 kurt=12.232 min=-0.4116 max=0.5125 XOM: mean=0.0029 sd=0.0384 skew=-0.105 kurt=6.657 min=-0.2007 max=0.1674

==========
EXOGENOUS INSTRUMENTS ==========
Count : 2 Names : Crude, SPY Rows matched : 536 (aligned with portfolio observations)

Exogenous empirical summary: Crude: mean=0.0031 sd=0.0576 skew=0.522 kurt=9.896 SPY: mean=0.0025 sd=0.0234 skew=-0.559 kurt=9.364

Portfolio x Exogenous historical correlations: Crude SPY AXP 0.175 0.729 BA 0.179 0.627 JPM 0.182 0.718 LMT 0.215 0.469 NGG 0.045 0.452 NVDA 0.131 0.633 PG 0.014 0.490 TSM 0.085 0.554 UAL 0.033 0.566 XOM 0.527 0.464

==========
CALIBRATING DISTRIBUTION PARAMETERS ==========
Distribution : rmvt Calibration method: FULL

Calibrating degrees of freedom: AXP: df = 8.5000 BA: df = 8.0000 JPM: df = 10.0000 LMT: df = 9.0000 NGG: df = 9.0000 NVDA: df = 10.0000 PG: df = 10.0000 TSM: df = 10.0000 UAL: df = 8.5000 XOM: df = 9.5000

Joint df (mean): 9.2500

==========
CALIBRATION SUMMARY ==========
Asset Optimal_DF Valid AXP 8.5 TRUE BA 8.0 TRUE JPM 10.0 TRUE LMT 9.0 TRUE NGG 9.0 TRUE NVDA 10.0 TRUE PG 10.0 TRUE TSM 10.0 TRUE UAL 8.5 TRUE XOM 9.5 TRUE

Joint df for simulation: 9.2500

rmvt: using calibrated df_joint = 9.2500

Reference asset defaulted to: AXP

==========
PORTFOLIO RISK SIMULATION (WEEKLY RETURNS) ==========
Assets : 10 — AXP, BA, JPM, LMT, NGG, NVDA, PG, TSM, UAL, XOM Reference asset : AXP (index 1) Observations : 536 Distribution : rmvt Copula : T Covariance estimation : GARCH Calibrate params : YES Propagation : YES `exogenous_shock` resolved from list (name format): SPY=-0.05, Crude=0.2 `exogenous_volatility_shock` resolved from list (name format): SPY=2, Crude=2

Weights: Equal weighted (1/N) Long-only portfolio

==========
WEIGHTS SUMMARY ==========
Portfolios : 1 Simulations each : 10,000 Total simulations : 10,000 Portfolio type : Long-Only

Portfolio weights: AXP : +0.1000 (+10.00%) BA : +0.1000 (+10.00%) JPM : +0.1000 (+10.00%) LMT : +0.1000 (+10.00%) NGG : +0.1000 (+10.00%) NVDA : +0.1000 (+10.00%) PG : +0.1000 (+10.00%) TSM : +0.1000 (+10.00%) UAL : +0.1000 (+10.00%) XOM : +0.1000 (+10.00%) SUM : +1.0000

==========
FITTING DISTRIBUTIONS ==========

Path: COPULA — marginals + copula separately Copula type: T AXP: t(=0.0041, _scale=0.0387, _marginal=0.0442, df=8.50) BA: t(=0.0031, _scale=0.0574, _marginal=0.0663, df=8.00) JPM: t(=0.0041, _scale=0.0335, _marginal=0.0375, df=10.00) LMT: t(=0.0030, _scale=0.0288, _marginal=0.0327, df=9.00) NGG: t(=0.0020, _scale=0.0264, _marginal=0.0300, df=9.00) NVDA: t(=0.0122, _scale=0.0570, _marginal=0.0637, df=10.00) PG: t(=0.0020, _scale=0.0218, _marginal=0.0244, df=10.00) TSM: t(=0.0066, _scale=0.0377, _marginal=0.0421, df=10.00) UAL: t(=0.0036, _scale=0.0629, _marginal=0.0720, df=8.50) XOM: t(=0.0029, _scale=0.0341, _marginal=0.0384, df=9.50)

==========
MARGINAL FITTING SUMMARY ==========
Distribution : rmvt Assets fitted: 10 Fitted means : 0.0041, 0.0031, 0.0041, 0.0030, 0.0020, 0.0122, 0.0020, 0.0066, 0.0036, 0.0029 Fitted SDs : 0.0387, 0.0574, 0.0335, 0.0288, 0.0264, 0.0570, 0.0218, 0.0377, 0.0629, 0.0341

Estimating GARCH forward volatilities… GARCH estimation complete

Forward-looking volatilities: AXP: 4.47% (historical: 3.87%) BA: 5.79% (historical: 5.74%) JPM: 3.44% (historical: 3.35%) LMT: 3.09% (historical: 2.88%) NGG: 3.79% (historical: 2.64%) NVDA: 5.68% (historical: 5.70%) PG: 3.48% (historical: 2.18%) TSM: 4.43% (historical: 3.77%) UAL: 6.84% (historical: 6.29%) XOM: 3.76% (historical: 3.41%)

Fitting copula for dependence structure… PIT method: fitted RMVT marginals Uniform range: [0.0000, 1.0000] n_obs=536 | n_params=46 | ratio=11.7 → strategy: unstructured + MPL Warm-start : [0.334, 0.949] (from Kendall's) t-copula correlation

not PD — applying nearPD t-copula fitted [unstructured + MPL]. df=8.00 | Log-lik: -2247116418577894635181592796646304459975882094073062457486468460039465759750356734507369482909585 97

Tail Dependence Coefficients: AXP — BA: =0.905, =0.521 AXP — JPM: =0.706, =0.244 AXP — LMT: =0.810, =0.356 AXP — NGG: =0.792, =0.333 AXP — NVDA: =0.793, =0.335 AXP — PG: =0.756, =0.292 AXP — TSM: =0.643, =0.195 AXP — UAL: =0.569, =0.150 AXP — XOM: =0.533, =0.132 BA — JPM: =0.676, =0.220 BA — LMT: =0.728, =0.264 BA — NGG: =0.803, =0.347 BA — NVDA: =0.709, =0.247 BA — PG: =0.467, =0.104 BA — TSM: =0.368, =0.072 BA — UAL: =0.657, =0.205 BA — XOM: =0.578, =0.155 JPM — LMT: =0.633, =0.189 JPM — NGG: =0.679, =0.222 JPM — NVDA: =0.909, =0.528 JPM — PG: =0.510, =0.122 JPM — TSM: =0.755, =0.291 JPM — UAL: =0.791, =0.332 JPM — XOM: =0.775, =0.314 LMT — NGG: =0.585, =0.159 LMT — NVDA: =0.695, =0.235 LMT — PG: =0.792, =0.334 LMT — TSM: =0.590, =0.162 LMT — UAL: =0.693, =0.233 LMT — XOM: =0.777, =0.316 NGG — NVDA: =0.849, =0.414 NGG — PG: =0.537, =0.134 NGG — TSM: =0.371, =0.073 NGG — UAL: =0.751, =0.287 NGG — XOM: =0.487, =0.112 NVDA — PG: =0.714, =0.251 NVDA — TSM: =0.758, =0.295 NVDA — UAL: =0.789, =0.330 NVDA — XOM: =0.703, =0.242 PG — TSM: =0.801, =0.345 PG — UAL: =0.421, =0.088 PG — XOM: =0.439, =0.094 TSM — UAL: =0.385, =0.077 TSM — XOM: =0.506, =0.120 UAL — XOM: =0.742, =0.278

Fitted copula df (8.00) differs from input copula_df (4).

Copula fit: Log-lik=-2247116418577894635181592796646304459975882094073062457486468460039465759750356 73
AIC=4494232837155789270363185593292608919951764188146124914972936920078931519500713469014738965819
BIC=4494232837155789270363185593292608919951764188146124914972936920078931519500713469014738965819

Final Σ not PD — applying nearPD

Using fitted marginal means (garch path)

rmvt scale matrix (df=9.25)

==========
STEP 6 DIAGNOSTICS ==========

Forward vs historical volatilities: AXP: 4.4697% ↑ (historical: 4.4209%) BA: 5.7898% ↓ (historical: 6.6252%) JPM: 3.4444% ↓ (historical: 3.7498%) LMT: 3.0867% ↓ (historical: 3.2660%) NGG: 3.7862% ↑ (historical: 2.9986%) NVDA: 5.6783% ↓ (historical: 6.3674%) PG: 3.4806% ↑ (historical: 2.4396%) TSM: 4.4345% ↑ (historical: 4.2106%) UAL: 6.8403% ↓ (historical: 7.1958%) XOM: 3.7598% ↓ (historical: 3.8391%)

Historical correlation: AXP BA JPM LMT NGG NVDA PG TSM UAL XOM AXP 1.0000 0.6270 0.7620 0.3850 0.3467 0.3557 0.2891 0.3257 0.6687 0.4834 BA 0.6270 1.0000 0.5658 0.4176 0.3767 0.3391 0.2643 0.3398 0.6417 0.4249 JPM 0.7620 0.5658 1.0000 0.3297 0.2515 0.3555 0.2872 0.3363 0.6526 0.4824 LMT 0.3850 0.4176 0.3297 1.0000 0.3190 0.2152 0.3297 0.1557 0.2505 0.4195 NGG 0.3467 0.3767 0.2515 0.3190 1.0000 0.1674 0.4355 0.1900 0.2512 0.2499 NVDA 0.3557 0.3391 0.3555 0.2152 0.1674 1.0000 0.1611 0.5696 0.2679 0.1275 PG 0.2891 0.2643 0.2872 0.3297 0.4355 0.1611 1.0000 0.1941 0.2494 0.1788 TSM 0.3257 0.3398 0.3363 0.1557 0.1900 0.5696 0.1941 1.0000 0.2881 0.1744 UAL 0.6687 0.6417 0.6526 0.2505 0.2512 0.2679 0.2494 0.2881 1.0000 0.3514 XOM 0.4834 0.4249 0.4824 0.4195 0.2499 0.1275 0.1788 0.1744 0.3514 1.0000

Copula correlation: AXP BA JPM LMT NGG NVDA PG TSM UAL XOM AXP 1.0000 0.9055 0.7059 0.8095 0.7916 0.7929 0.7560 0.6430 0.5691 0.5335 BA 0.9055 1.0000 0.6763 0.7276 0.8030

0.7093 0.4668 0.3675 0.6569 0.5780 JPM 0.7059 0.6763 1.0000 0.6330 0.6793 0.9086 0.5102 0.7548 0.7910 0.7752 LMT 0.8095 0.7276 0.6330 1.0000 0.5850 0.6951 0.7925 0.5904 0.6926 0.7774 NGG 0.7916 0.8030 0.6793 0.5850 1.0000 0.8494 0.5372 0.3715 0.7506 0.4873 NVDA 0.7929 0.7093 0.9086 0.6951 0.8494 1.0000 0.7135 0.7582 0.7890 0.7035 PG 0.7560 0.4668 0.5102 0.7925 0.5372 0.7135 1.0000 0.8012 0.4207 0.4392 TSM 0.6430 0.3675 0.7548 0.5904 0.3715 0.7582 0.8012 1.0000 0.3853 0.5062 UAL 0.5691 0.6569 0.7910 0.6926 0.7506 0.7890 0.4207 0.3853 1.0000 0.7418 XOM 0.5335 0.5780 0.7752 0.7774 0.4873 0.7035 0.4392 0.5062 0.7418 1.0000

Difference (Copula − Historical): AXP BA JPM LMT NGG NVDA PG TSM UAL XOM AXP 0.0000 0.2785 -0.0560 0.4245 0.4450 0.4373 0.4669 0.3173 -0.0995 0.0501 BA 0.2785 0.0000 0.1105 0.3100 0.4264 0.3702 0.2025 0.0277 0.0152 0.1531 JPM -0.0560 0.1105 0.0000 0.3033 0.4278 0.5531 0.2229 0.4185 0.1384 0.2927 LMT 0.4245 0.3100 0.3033 0.0000 0.2660 0.4799 0.4628 0.4347 0.4421 0.3579 NGG 0.4450 0.4264 0.4278 0.2660 0.0000 0.6820 0.1018 0.1815 0.4995 0.2374 NVDA 0.4373 0.3702 0.5531 0.4799 0.6820 0.0000 0.5524 0.1886 0.5211 0.5760 PG 0.4669 0.2025 0.2229 0.4628 0.1018 0.5524 0.0000 0.6070 0.1714 0.2604 TSM 0.3173 0.0277 0.4185 0.4347 0.1815 0.1886 0.6070 0.0000 0.0972 0.3318 UAL -0.0995 0.0152 0.1384 0.4421 0.4995 0.5211 0.1714 0.0972 0.0000 0.3903 XOM 0.0501 0.1531 0.2927 0.3579 0.2374 0.5760 0.2604 0.3318 0.3903 0.0000 Max |diff|: 0.6820

Final Σ: AXP BA JPM LMT NGG NVDA PG TSM UAL XOM AXP 0.001998 0.002343 0.001087 0.001117 0.001340 0.002012 0.001176 0.001274 0.001740 0.000896 BA 0.002343 0.003352 0.001349 0.001300 0.001760 0.002332 0.000941 0.000944 0.002602 0.001258 JPM 0.001087 0.001349 0.001186 0.000673 0.000886 0.001777 0.000612 0.001153 0.001864 0.001004 LMT 0.001117 0.001300 0.000673 0.000953 0.000684 0.001218 0.000851 0.000808 0.001462 0.000902 NGG 0.001340 0.001760 0.000886 0.000684 0.001433 0.001826 0.000708 0.000624 0.001944 0.000694 NVDA 0.002012 0.002332 0.001777 0.001218 0.001826 0.003224 0.001410 0.001909 0.003065 0.001502 PG 0.001176 0.000941 0.000612 0.000851 0.000708 0.001410 0.001211 0.001237 0.001002 0.000575 TSM 0.001274 0.000944 0.001153 0.000808 0.000624 0.001909 0.001237 0.001966 0.001169 0.000844 UAL 0.001740 0.002602 0.001864 0.001462 0.001944 0.003065 0.001002 0.001169 0.004679 0.001908 XOM 0.000896 0.001258 0.001004 0.000902 0.000694 0.001502 0.000575 0.000844 0.001908 0.001414

Condition number : 145033954.83 Min eigenvalue : 1.54e-10

Uniformity diagnostic (KS test): Asset KS_stat p_value Flag AXP 0.0660 0.0188 REJECT BA 0.0976 0.0001 REJECT JPM 0.0413 0.3187 OK LMT 0.0590 0.0479 REJECT NGG 0.0516 0.1156 OK NVDA 0.0343 0.5553 OK PG 0.0496 0.1436 OK TSM 0.0359 0.4930 OK UAL 0.0698 0.0108 REJECT XOM 0.0626 0.0300 REJECT

Tail dependence retrieved from Step 5 (df=8.0000)

Pair _copula
AXP — BA 0.9055 0.5208 HIGH
AXP — JPM 0.7059 0.2444
AXP — LMT 0.8095 0.3558 HIGH
AXP — NGG 0.7916 0.3330 HIGH
AXP — NVDA 0.7929 0.3346 HIGH
AXP — PG 0.7560 0.2924
AXP — TSM 0.6430 0.1955
AXP — UAL 0.5691 0.1504
AXP — XOM 0.5335 0.1324
BA — JPM 0.6763 0.2199

Pair _copula
BA — LMT 0.7276 0.2640
BA — NGG 0.8030 0.3473 HIGH
BA — NVDA 0.7093 0.2473
BA — PG 0.4668 0.1039
BA — TSM 0.3675 0.0717
BA — UAL 0.6569 0.2053
BA — XOM 0.5780 0.1552
JPM — LMT 0.6330 0.1887
JPM — NGG 0.6793 0.2223
JPM — NVDA 0.9086 0.5280 HIGH
JPM — PG 0.5102 0.1217
JPM — TSM 0.7548 0.2912
JPM — UAL 0.7910 0.3322 HIGH
JPM — XOM 0.7752 0.3135 HIGH
LMT — NGG 0.5850 0.1591
LMT — NVDA 0.6951 0.2352
LMT — PG 0.7925 0.3340 HIGH
LMT — TSM 0.5904 0.1622
LMT — UAL 0.6926 0.2331
LMT — XOM 0.7774 0.3160 HIGH
NGG — NVDA 0.8494 0.4141 HIGH
NGG — PG 0.5372 0.1342
NGG — TSM 0.3715 0.0728
NGG — UAL 0.7506 0.2868
NGG — XOM 0.4873 0.1120
NVDA — PG 0.7135 0.2511
NVDA — TSM 0.7582 0.2947
NVDA — UAL 0.7890 0.3297 HIGH
NVDA — XOM 0.7035 0.2422
PG — TSM 0.8012 0.3449 HIGH
PG — UAL 0.4207 0.0877
PG — XOM 0.4392 0.0939
TSM — UAL 0.3853 0.0767
TSM — XOM 0.5062 0.1200
UAL — XOM 0.7418 0.2778

Max |latent_cor − copula_cor|: 0.6893

base_sigma not PD before stress testing — applying nearPD

Final Σ condition number : 145033956.50 Final Σ min eigenvalue : 1.54e-10

==========
STRESS TESTING ==========
Portfolio return-shocked : 0 — none Portfolio vol-shocked : 0 — none Exogenous return-shocked :
2 — SPY, Crude Exogenous vol-shocked : 2 — SPY, Crude Propagation : ENABLED

— EXOGENOUS VOLATILITY SHOCK — Exogenous Asset Multiplier Crude 2.0000x SPY 2.0000x Exogenous covariance blocks scaled

— RETURN SHOCK & PROPAGATION —

Exogenous return shocks (zero portfolio weight — not simulated): SPY: -0.0500 (-5.00%) Crude: +0.2000 (+20.00%)

Conditional mean propagation: Asset Base Stressed Δ Type ——————————————————————————— AXP 0.00408 -0.03013 -0.03422 [propagated] BA 0.00313 -0.03699 -0.04012 [propagated] JPM 0.00412 -0.02373 -0.02785 [propagated] LMT 0.00303 -0.00614 -0.00917 [propagated] NGG 0.00199 -0.01641 -0.01840 [propagated] NVDA 0.01217 -0.03334 -0.04551 [propagated] PG 0.00196 -0.01602 -0.01798 [propagated] TSM 0.00656 -0.02244 -0.02900 [propagated] UAL 0.00356 -0.05519 -0.05875 [propagated] XOM 0.00286 0.01758 0.01472 [propagated] Exogenous (not in portfolio, shown for reference): SPY 0.00250 -0.04750 -0.05000 [EXOG SHOCKED] Crude 0.00313 0.20313 0.20000 [EXOG SHOCKED]

— EXOGENOUS VOLATILITY ADDON (LTV) — Crude: k_e=2.00 → (k_e^2−1)=3.0000 × outer product added SPY: k_e=2.00 → (k_e^2−1)=3.0000 × outer product added Asset marginal vols: pre=4.470%, 5.790%, 3.444%, 3.087%, 3.786%, 5.678%, 3.481%, 4.434%, 6.840%, 3.760% | post=7.277%, 9.462%, 5.918%, 4.247%, 4.462%, 9.110%, 4.049%, 6.031%, 9.830%, 5.991% sigma_stressed not PD — applying nearPD

==========
STRESSED PARAMETERS SUMMARY ==========

Asset _base _stressed Δ
AXP 0.00408 -0.03013 -0.03422 [propagated]
BA 0.00313 -0.03699 -0.04012 [propagated]
JPM 0.00412 -0.02373 -0.02785 [propagated]
LMT 0.00303 -0.00614 -0.00917 [propagated]
NGG 0.00199 -0.01641 -0.01840 [propagated]
NVDA 0.01217 -0.03334 -0.04551 [propagated]
PG 0.00196 -0.01602 -0.01798 [propagated]
TSM 0.00656 -0.02244 -0.02900 [propagated]
UAL 0.00356 -0.05519 -0.05875 [propagated]
XOM 0.00286 0.01758 0.01472 [propagated]

Asset _base _stressed Δ
AXP 4.4697 7.2768 2.8071 [propagated]
BA 5.7898 9.4617 3.6719 [propagated]
JPM 3.4444 5.9176 2.4732 [propagated]
LMT 3.0867 4.2465 1.1598 [propagated]
NGG 3.7862 4.4624 0.6762 [propagated]
NVDA 5.6783 9.1102 3.4319 [propagated]
PG 3.4806 4.0494 0.5688 [propagated]
TSM 4.4345 6.0315 1.5970 [propagated]
UAL 6.8403 9.8304 2.9901 [propagated]
XOM 3.7598 5.9913 2.2315 [propagated]

Asset	_base	_stressed	Δ

Exogenous shock applied (zero weight — not simulated): SPY Δret=-0.0500 vol_mult=2.0000x Crude Δret=+0.2000 vol_mult=2.0000x

sigma_stressed condition number : 146636201.75 sigma_stressed min eigenvalue : 4.11e-10

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SIMULATING PORTFOLIO RETURNS ==========
base_sigma not PD — applying nearPD sigma_stressed not PD — applying nearPD rmvt: deflating covariance to scale matrix (df=9.2500) rmvt df: 9.2500

Simulating 10,000 scenarios across 1 weight set(s)…

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SIMULATION SUMMARY ==========
Distribution : rmvt Simulation df : 9.2500 Weight sets : 1 Sims per set : 10,000 Valid pre-obs : 10,000 / 10,000 Valid post-obs : 10,000 / 10,000

Sanity check: Portfolio Mean Target: 0.4347% Simulated: 0.4750% Portfolio Target: 3.7547% Simulated: 3.7340% Simulated vol consistent with target

==========
PRE-STRESS PORTFOLIO STATISTICS (Weekly Returns) ==========

Sample size: 10,000

RETURN (Weekly) Expected Return (Mean) : +0.4750% [primary] Expected Return (median) : +0.4875%

RETURN (Annualised) Annual Return (geometric) : +27.9404% Annual Volatility : 26.9263% Sharpe Ratio (annual) : 0.9172

RISK (Weekly) Volatility : 3.7340% Skewness : -0.0314 Excess Kurtosis : 1.0978

VALUE-AT-RISK (Weekly, negative = loss) 90% VaR (10th pct) : -4.0842% 95% VaR (5th pct) : -5.5452% 99% VaR (1st pct) : -8.7100%

EXPECTED SHORTFALL (Weekly) 90% CVaR : -6.1443% 95% CVaR : -7.5474% 99% CVaR : -10.9956%

TAIL DIAGNOSTICS % Positive months : 55.33% % Negative months : 44.67% Worst single return : -22.9172% Best single return : +18.1171%

==========
POST-STRESS PORTFOLIO STATISTICS (Weekly Returns) ==========

Sample size: 10,000

RETURN (Weekly) Expected Return (Mean) : -2.2075% [primary] Expected Return (median) : -2.1767%

RETURN (Annualised) Annual Return (geometric) : -68.6748% Annual Volatility : 44.1035% Sharpe Ratio (annual) : -2.6027

RISK (Weekly) Volatility : 6.1161% Skewness : 0.0529 Excess Kurtosis : 1.4258

VALUE-AT-RISK (Weekly, negative = loss) 90% VaR (10th pct) : -9.6795% 95% VaR (5th pct) : -12.1374% 99% VaR (1st pct) : -17.1416%

EXPECTED SHORTFALL (Weekly) 90% CVaR : -13.1069% 95% CVaR : -15.3970% 99% CVaR : -20.5241%

TAIL DIAGNOSTICS % Positive months : 34.25% % Negative months : 65.75% Worst single return : -36.5645% Best single return : +43.2543%

==========
STRESS IMPACT SUMMARY ==========

Metric	Pre-Stress	Post-Stress	Change
Exp Return (Mean)	0.4750%	-2.2075%	-2.6824%
Volatility	3.7340%	6.1161%	+2.3821%
Skewness	-0.0314	0.0529	+0.0843
Excess Kurtosis	1.0978	1.4258	+0.3280
95% VaR	-5.5452%	-12.1374%	-6.5922%
99% VaR	-8.7100%	-17.1416%	-8.4316%
95% CVaR	-7.5474%	-15.3970%	-7.8496%

Warning: VaR95 worsened by 6.59% Warning: CVaR95 worsened by 7.85% Warning: Vol increased by 2.38%

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GENERATING PORTFOLIO DIAGNOSTIC PLOTS ==========
Plot 1: Asset distributions Plot 2: Portfolio distribution Plot 3: QQ plot Plot 4: Historical correlation Plot 5: Copula correlation + difference Plot 6: Covariance matrix

Building GARCH conditional volatility time-series plots… Plot 7: GARCH time-series (10 asset(s), 4 col × 3 row canvas)

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GENERATING PAIRWISE COPULA DIAGNOSTIC PLOTS Reference asset: AXP ==========
Pairs: AXP vs BA, JPM, LMT, NGG, NVDA, PG, TSM, UAL, XOM Plot 11: Pairwise diagnostics (9 pairs, 4-column layout)

Building tail dependency & correlation divergence plots… Plot 11b: Tail dependency & correlation divergence

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CREATING STRESS PROPAGATION PLOTS ==========
Scenario : B: Exogenous (non-portfolio instruments) Exog shocked (2) : SPY, Crude Plot 12b: Scenario B — exogenous-only propagation

Building pre/post stress distribution comparison… Plot 12c: Pre/post distribution comparison

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COMPILING RESULTS ==========

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SIMULATION COMPLETE ==========
Pre-stress samples : 10,000 Post-stress samples : 10,000 Diagnostic plots : 10 Stress plots : 2